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EXPERIMENT – 26

MATLAB SIMULATION OF SINGLE-BIT ERROR DETECTION AND CORRECTION USING SYNDROME VECTORS IN LINEAR BLOCK CODES

PROGRAM:

```
clc;
clear;
close all;
% (7,4) Hamming Code
msg = [1 0 1 1];
% Encode
code = encode(msg,7,4,'hamming/binary');
% Introduce Single-Bit Error
rx = code;
rx(3) = ~rx(3); % Flip one bit
% Decode and Correct Error
dec = decode(rx,7,4,'hamming/binary');
% Display Results
disp('Message Bits:')
disp(msg)
disp('Encoded Bits:')
disp(code)
disp('Received Bits (with error):')
disp(rx)
disp('Corrected Message:')
disp(dec)
% Plot
figure;
subplot(2,1,1)
```

```
stem(code,'filled')
title('Encoded Codeword')
grid on
subplot(2,1,2)
stem(rx,'filled')
title('Received Codeword with Single-Bit Error')
grid on
```

OUTPUT:

